LAB 6

Code and Branch Coverage Analysis

Software Engineering Lab

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GitHub repository: github.com/chaithanya2810/SE-Labs-PES1UG24CS124

1. Objective

The objective of this lab was to understand code coverage and branch coverage, use the coverage tool to measure test coverage, and add test cases for the paths that were not covered by the starter tests.

2. Application under test

The Python module credit_card_rewards.py contains three functions:

- reward_points() - calculates reward points using the card tier, amount, spending category and weekend status.
- late_payment_fee() - calculates the fee based on the outstanding amount and the number of days late.
- card_status() - returns the account status using utilisation, missed payments and frozen-card information.

3. Test approach

I used pytest to run the tests and the Python coverage package to measure coverage. I kept the application code unchanged and added two separate test files:

- CodeCoverageTest_PES1UG24CS124_application.py - contains 24 tests for the application functions, including normal inputs and boundary or invalid inputs.
- BranchCoverageTest_PES1UG24CS124_application.py - contains 28 tests to exercise the different outcomes of the decision statements.

4. Coverage concepts

Line coverage	Measures how many executable lines were run at least once during testing.
Branch coverage	Measures whether the possible outcomes of decision statements, such as if/else branches, were exercised.

5. Coverage results

The following screenshots show the final test runs and the coverage reports.

5.1 Code coverage

```
PS D:\SE-Lab-Work\SE-Labs-PES1UG24CS124\Lab6\L6_L7_61_credit_card_rew
ards\LAB6> python -m coverage erase
PS D:\SE-Lab-Work\SE-Labs-PES1UG24CS124\Lab6\L6_L7_61_credit_card_rew
ards\LAB6> python -m coverage run --source=credit_card_rewards -m pyt
est -q CodeCoverageTest_PES1UG24CS124_application.py
.......................                                  [100%]
24 passed in 0.21s
PS D:\SE-Lab-Work\SE-Labs-PES1UG24CS124\Lab6\L6_L7_61_credit_card_rew
ards\LAB6> python -m coverage report -m
Name                    Stmts   Miss  Cover   Missing
-------------------------------------------------------
credit_card_rewards.py      35      0   100%
-------------------------------------------------------
TOTAL                       35      0   100%
PS D:\SE-Lab-Work\SE-Labs-PES1UG24CS124\Lab6\L6_L7_61_credit_card_rew
ards\LAB6>
```

Figure 1. Code coverage test run: 24 tests passed and 100% line coverage.

5.2 Branch coverage

```
PS D:\SE-Lab-Work\SE-Labs-PES1UG24CS124\Lab6\L6_L7_61_credit_card_reward
s\LAB6> python -m coverage erase
PS D:\SE-Lab-Work\SE-Labs-PES1UG24CS124\Lab6\L6_L7_61_credit_card_reward
s\LAB6> python -m coverage run --branch --source=credit_card_rewards -m
pytest -q BranchCoverageTest_PES1UG24CS124_application.py
............................                             [100%]
28 passed in 0.35s
PS D:\SE-Lab-Work\SE-Labs-PES1UG24CS124\Lab6\L6_L7_61_credit_card_reward
s\LAB6> python -m coverage report -m
Name                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------
credit_card_rewards.py      35      0     26      0   100%
-----------------------------------------------------------------------
TOTAL                       35      0     26      0   100%
PS D:\SE-Lab-Work\SE-Labs-PES1UG24CS124\Lab6\L6_L7_61_credit_card_reward
s\LAB6>
```

Figure 2. Branch coverage test run: 28 tests passed; 35 statements and 26 branches covered, with 0 partial branches.

6. Result and conclusion

The code coverage test run passed all 24 tests and reported 100% line coverage (35 statements, 0 missed). The branch coverage test run passed all 28 tests and reported 100% coverage, with 26 branches and 0 partially covered branches. The added tests exercised the statements and decision branches measured by the coverage tool for this module.

7. Git commits

The two commits pushed to the personal repository were 15c9f23 (code coverage tests) and 0594617 (branch coverage tests).